Jisoo Jang

Deep thinking. Fast execution. Continuous improvement.

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Education

2022–2026 **Doctor of Philosophy in Data Science**, *Graduate School of Data Science, Seoul National University*, Seoul.
Large Language Models (LLMs) and Context Engineering, LLM-Driven Natural Language Interfaces for Databases, Natural Langauge to Structured Query Language (NL2SQL)

2020–2021 : **Master in Data Science**, *Graduate School of Data Science, Seoul National University*, Seoul.

2010–2018 : **BA in Economics & Finance**, *College of Economics and Finance, Hanyang University*, Seoul.

Publications

Journal Articles

2026 Jisoo Jang, Mingyoo Son, and Wen-Syan Li. Nl2sql+: An extended framework for sql generation with on-demand embedded ml functions. *IEEE Access*, volume 14, pages 126645–126660, 2026.

In Conference Proceedings

2026 Joohyun Lee, Ghita Benboubker, JungKwan Han, Jisoo Jang, and Wen-Syan Li. Pathfinder: Context engineering and knowledge management for domain-specific horizontal reasoning. In *2026 IEEE 42nd International Conference on Data Engineering (ICDE)*, pages 4223–4226, 2026.

2025 Jisoo Jang and Wen-Syan Li. Retrieval-augmented nl2sql generation with data-centric query capsules for enterprise applications. In *Proceedings of the 2025 Annual International ACM SIGIR Conference on Research and Development in Information Retrieval in the Asia Pacific Region*, pages 123–132, 2025.

2024 Jisoo Jang and Wen-Syan Li. Au-rag: agent-based universal retrieval augmented generation. In *Proceedings of the 2024 Annual International ACM SIGIR Conference on Research and Development in Information Retrieval in the Asia Pacific Region*, pages 2–11, 2024.

2023 Jisoo Jang, Mina Kim, Tien-Cuong Bui, and Wen-Syan Li. Toward interpretable machine learning: Constructing polynomial models based on feature interaction trees. In *Pacific-Asia Conference on Knowledge Discovery and Data Mining*, pages 159–170. Springer, 2023.

arXiv Papers

2026 Jisoo Jang Wen-Syan Li. Twinbi: An agentic digital twin for efficient augmented interactions with business intelligence dashboards. *arXiv preprint arXiv:2606.13731*, 2026.

2025 Jisoo Jang, Tien-Cuong Bui, Yunjun Choi, and Wen-Syan Li. Prompt tuning for natural language to sql with embedding fine-tuning and rag. *arXiv preprint arXiv:2511.08245*, 2025.

Under Review

2026 Jisoo Jang, Mingyoo Son, and Wen-Syan Li, When Should a Routing Decision Be Re-examined? A Diagnostic Study of Model Switching During Reasoning, In NeurIPS 2026.

2026 Jisoo Jang, Mingyoo Son, and Wen-Syan Li, SQLSense: Unifying Generation, Routing, and Repair in Small Text-to-SQL Models, In AAAI 2027.

2026 Jisoo Jang and Wen-Syan Li, AgentScope: Evolving Reusable Agent Pools to Meet User Wish List with Validation, In AAAI 2027.

2026 Jisoo Jang, Ghita Benboubker and Wen-Syan Li, State-Grounded Conversational Retrieval for Interactive Analytical Dashboards, In SIGIR-AP 2026.

Projects

2024.11 – 2025.01 (3M) ***Financial Q&A Chatbot Based on Financial Reports***.
- Building a financial Q&A chatbot based on financial reports in collaboration between SNU KDT and Naver Cloud
- Led a team by helping define the problem, design experiments, and manage the project through weekly meetings
- Explained concepts of LLM and RAG, and guided system architecture design by applying the LLM Compiler paper

2021.04 – 2021.06 (2M) ***Metric for Deep Learning Visualization Technique***.
- Proposed I-metric, a metric to evaluate neuron visualization as part of Explainable AI techniques
- Conducted experiments to verify whether an additional tool (grid-spec activation camera) improves evaluation
- Achieved a 28% improvement over baseline visualization when using the additional tool — paper link
- Role: Implemented the grid-spec activation camera

2020.11 – 2021.11 (12M) ***Improving Cleanness of Dishwasher***.
- A project with SK Networks to improve dishwasher cleaning performance using pattern clustering on turbidity sensor data

2020.10 - 2021.02 (5M) ***Battery Life Expectation***.
- Predicted the remaining battery life of a logistics container tracking device by solving it as a regression problem for voltage values

Work Experience

2022.03 - 2026.08 (4.5Y) ***Research***, *Applied Machine Learning Lab*, Seoul.
- Conducting doctoral research on large language models, context engineering, intelligent agents, and natural language interfaces for enterprise databases
  - Developed retrieval-augmented NL2SQL methods, including agent-based universal RAG, data-centric query capsules, and embedding fine-tuning
  - Proposed Strategy Map, an inference-time process guidance layer that maintains completed, available, and blocked workflow states for dependency-sensitive agents
  - Improved task accuracy by 22.9 percentage points with ReAct and 38.6 percentage points with a tool-retrieval-based agent architecture on enterprise workflow tasks
- Published research at PAKDD 2023, SIGIR-AP 2024 and 2025, IEEE ICDE 2026(Demo Track), and IEEE Access
- Led applied LLM research projects
  - Led the development of a financial-report Q&A chatbot in collaboration with SNU KDT and Naver Cloud
  - Defined research problems, designed experiments, guided LLM and RAG system architecture, and managed weekly project meetings

2020.03 - 2022.02 (2Y) ***Research***, *Applied Machine Learning Lab*, Seoul.
- Conducted master's research on natural language interfaces for databases, task-oriented dialogue systems, and applied machine learning
  - Developed a financial question-answering framework combining BERT-based natural language understanding, financial-statement ontology, Text-to-SQL, and embedded machine learning
  - Implemented an ontology-based dialogue manager that translated financial questions into database queries and supported predictive what-if analysis
- Conducted applied machine learning research across explainable AI and industrial sensor-data applications
  - Proposed I-metric for evaluating neuron-visualization methods and implemented a grid-spec activation camera, achieving a 28% improvement over the baseline visualization
  - Analyzed turbidity sensor patterns in a 12-month collaboration with SK Networks to improve dishwasher cleaning performance
  - Developed a regression-based method for predicting the remaining battery life of logistics container tracking devices

2018.02 - 2019.08 (1.5Y) ***Boostcourse AI***, *Naver Connect*, Seoul.

- Publishing AI-related content on edwith (changed to boostcourse) platform to make it easier for learners to study
  - Professor KyungHyun Cho's course: Natural Language Processing with Deep Learning
  - Professor Andrew Ng's Deep Learning series: 1, 2, 3, 4
- Planning and managing project-based learning content
  - Assisted in planning and managing the course Deep Learning for Everyone Season 2
  - Planned, set up the learning environment, and reviewed code for "Introduction to Deep Learning with TensorFlow and PyTorch"

Position of Responsibility

2025 Volunteer for conference, SIGIR-AP 2025.

2024 Student Helper to manage the in-person part, PAKDD 2024 Robust Machine Learning in Open Environments Workshop.

Awards

2025.10 SIGIR Student Travel Grant Award

2024.10 SIGIR Student Travel Grant Award

2023.08 The 11th Ministry of Trade, Industry, and Energy Public Data Utilization BI Contest - 3rd prize in the "Public Domain" Area

Lectures

2024.08 A Guide to Building LLM-Based Applications: Prompt Engineering and RAG.
@ Jeonju IT & CT Industry Promotion Agency

2022.07 Introduction to DeepLearning with PyTorch.
@ University of Seoul

2021.07 Python & PyTorch Basic.
@ HanKyong University

2019.12 Introduction to DeepLearning with PyTorch.
@ Multicampus

Teaching Assistantship

Fall, 2022 – Present M3239.006700: Project for Data Science, GSDS, SNU.

Fall, 2023 – Spring, 2025 M3239.002300: Prompt Engineering: Art and Science for Interactions with LLM, GSDS, SNU.

Fall, 2022 – Spring, 2023 M3239.005100: Big Data and Knowledge Management Systems 2, GSDS, SNU.

Publication (Book)

2019.06 PyTorch for Deep Learning

Interests

Attendances for Learning New Knowledge

2025.12 Attend the 3rd International ACM SIGIR Conference on Information Retrieval in the Asia Pacific, Xi'an, China

2025.08 Attend Agentic AI Summit 2025 in UC Berkeley, California, USA

2024.12 Attend the 2nd International ACM SIGIR Conference on Information Retrieval in the Asia Pacific, Tokyo, Japan

2024.05 Attend the 28th Pacific-Asia Conference on Knowledge Discovery and Data Mining, Taipei, Taiwan

2023.05 Attend the 27th Pacific-Asia Conference on Knowledge Discovery and Data Mining, Osaka, Japan

2022.12 Attend 2022 IEEE International Conference on Big Data, Tokyo, Japan

2018.08 Attend Pycon-Korea 2018, Seoul, Korea

2018.01 Participate "Deep Learning Basic with Pytorch" Class @ FastCampus, Seoul, Korea

2017.10 Participate "Text Mining using Python" Class @ FastCampus, Seoul, Korea

2017.08 Attend Pycon-Korea 2017, Seoul, Korea

2017.05 Learned ML @ FastCampus Data Science School, Seoul, Korea

Participate or Running A Study Group

2022.11 (3M) Head first Rust! @ Modulabs

2020.05 (3M) Participate Beyond BERT @ Modulabs: reviewed Fast-BERT

2018.08 (9M) Ran a paper study with Yonsei University Master course students (Learning NLP) Drive Link